

Multiple Sequence Alignment (MSA)

Chapter 12 &
Notredame 2002.pdf

1. Uses of MSA

2. Technical difficulties

1. Select sequences
2. Select objective function
3. Optimize the objective function
 1. Exact algorithms
 2. Progressive algorithms
 3. Iterative algorithms
 4. Consistency-based algorithms

3. Tools to view alignments

1. MEGA
2. BOXSHADE & Seq. LOGOS

USES of Multiple Sequence Alignment

- **Sequence relationships**
- **Conservation patterns**
 - Mutable regions
 - Conserved residues
 - Conserved properties
 - Conserved sequence patterns
 - Domain boundaries
 - [...]



	56	63	96
	NLVD	CVSE...ND.G	CGGGY ... SCM ...
	ELVD	CER...RSH.G	CKGGY ... TCR ...
	ELVE	CTNG.QNS.G	CNGGL ... KCD ...
	ELVD	CDR.S.YNE.G	CDGGL ... VCD ...
CYSL_LYCES	...	ELVD	CDR.S.YNE.G
CYSP_HEMSP	...	ELVD	CDKEE...NQ.G
CATL_DROME	...	NLVD	CDST.KYGNN.G
CATJ_RAT	...	NLLDTKSE...	GI.GLPWGT ... PCR ...
ALEU_HORVU	...	QLVD	CAG.GFNNF.G
BROM_ANACO	...	QVLDCAK...	GY.GCKGGW ... TCK ...
EUM1_EURMA	...	ELVD	CAS...QN.G
CPR5_CAEEL	...	DLLS	CCTGMFSCGNG
CYS1_OSTOS	...	DVVS	CCTWCGD...G

MSA can be used to:

- **Infer function**
- **Predict secondary structure**
- **Phylogenetic reconstruction**
- **Identify residues important for function**
- **Sensitive database searching algorithms**

PSI-BLAST

Position Specific Iterated - BLAST

Begin with BLAST search
Identify significant hits
Align to query
Compile into position specific scoring matrix (PSSM or "sequence profile")
Repeat search with profile
Add new aligned hits to PSSM
Iterate until no new sequences can be added

If the MSA is incorrect, all the above inferences will be incorrect!

PSI-BLAST (Position Specific Iterative BLAST)

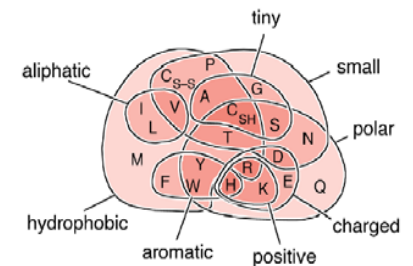
- Designed to detect weak relationships
- The added sensitivity comes from the use of a **profile** that is constructed (automatically) from a multiple alignment.
- The profile is generated by calculating a **Position-Specific Scoring Matrix (PSSM)** for every position in the alignment. Also called profiles of Hidden Markov Models
- **PSSM** are numerical representations of a multiple alignment
- A highly conserved position receives a high score.
- The profile is used to perform additional searches (iteration) and the results of each iteration used to refine the profile.
- Each iteration uses a PSSM built from the previous iteration.
- Continue search iteratively until no new matches are identified: "convergence".

APHIIVATPG
GCEIVATPG
GVEICATPG
GVDILIGT
RPHIIVATPG
KPHIIVATPG
KVQLIATPG
RPDIVATPG
APHIIVGTPG
APHIIVGTPG
GCHVVATPG
NQDIVVATPG

Construction of
a **PSSM**

PSI-BLAST steps

- BLASTP
- Multiple Alignment
- Construct PSSM
- Use PSSM to search



Cons	A	B	C	D	E	F	G	H	I	K	L	M	N	P	Q	R	S	T	V	W	Y	Z
G	17	18	0	19	14	-22	31	0	-9	12	-15	-5	15	10	9	6	18	14	1	-15	-22	11
P	18	0	13	0	0	-12	13	0	8	-3	-3	-1	-2	23	2	-2	12	11	17	-31	-8	1
H	5	24	-12	29	25	-20	8	32	-9	9	-10	-9	22	7	30	10	0	4	-8	-20	-7	27
I	-1	-12	6	-13	-11	33	-12	-13	63	-11	40	29	-15	-9	-14	-15	-6	7	50	-17	8	-11
V	3	-11	1	-11	-9	22	-3	-11	46	-9	37	30	-13	-3	-9	-13	-6	6	50	-19	2	-8
V	5	-9	9	-9	-9	19	-1	-13	57	-9	35	26	-13	-2	-11	-13	-4	9	58	-29	0	-9
A	54	15	12	20	17	-24	44	-6	-4	-1	-11	-5	12	19	9	-13	21	19	9	-39	-20	10
T	40	20	20	20	20	-30	40	-10	20	20	-10	0	20	30	-10	-10	30	150	20	-60	-30	10
P	31	6	7	6	6	-41	19	11	-9	6	-16	-11	0	89	17	17	24	22	9	-50	-48	12
G	70	60	20	70	50	-60	150	-20	-30	-10	-50	-30	40	30	20	-30	60	40	20	-100	-70	30

Each columns in the alignment is a row in the PSSM

Frequency of occurrence of a residue at each position

Calculate Pb of each aa at each position

T at position 8 conserved= highest score 150

P at position 9 less conserve= score 89

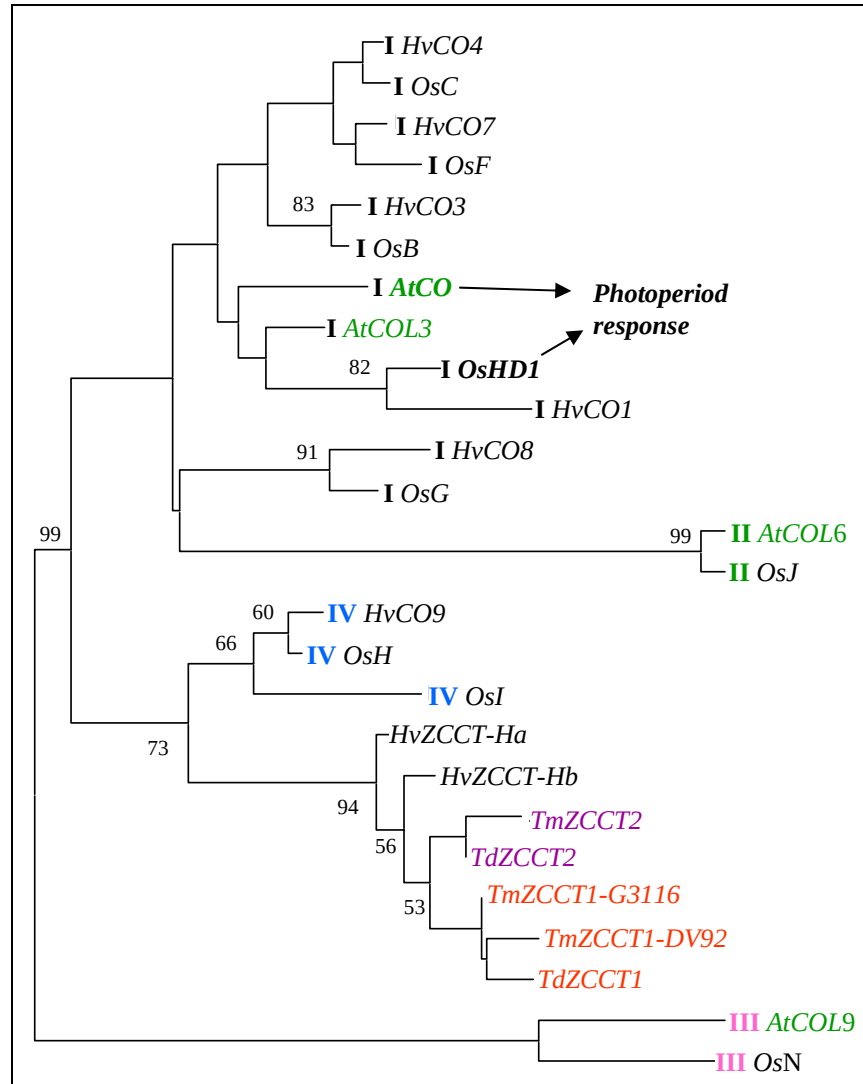
Note low scores of aromatic FYW relative to A at P row

MSA of ZCCT genes

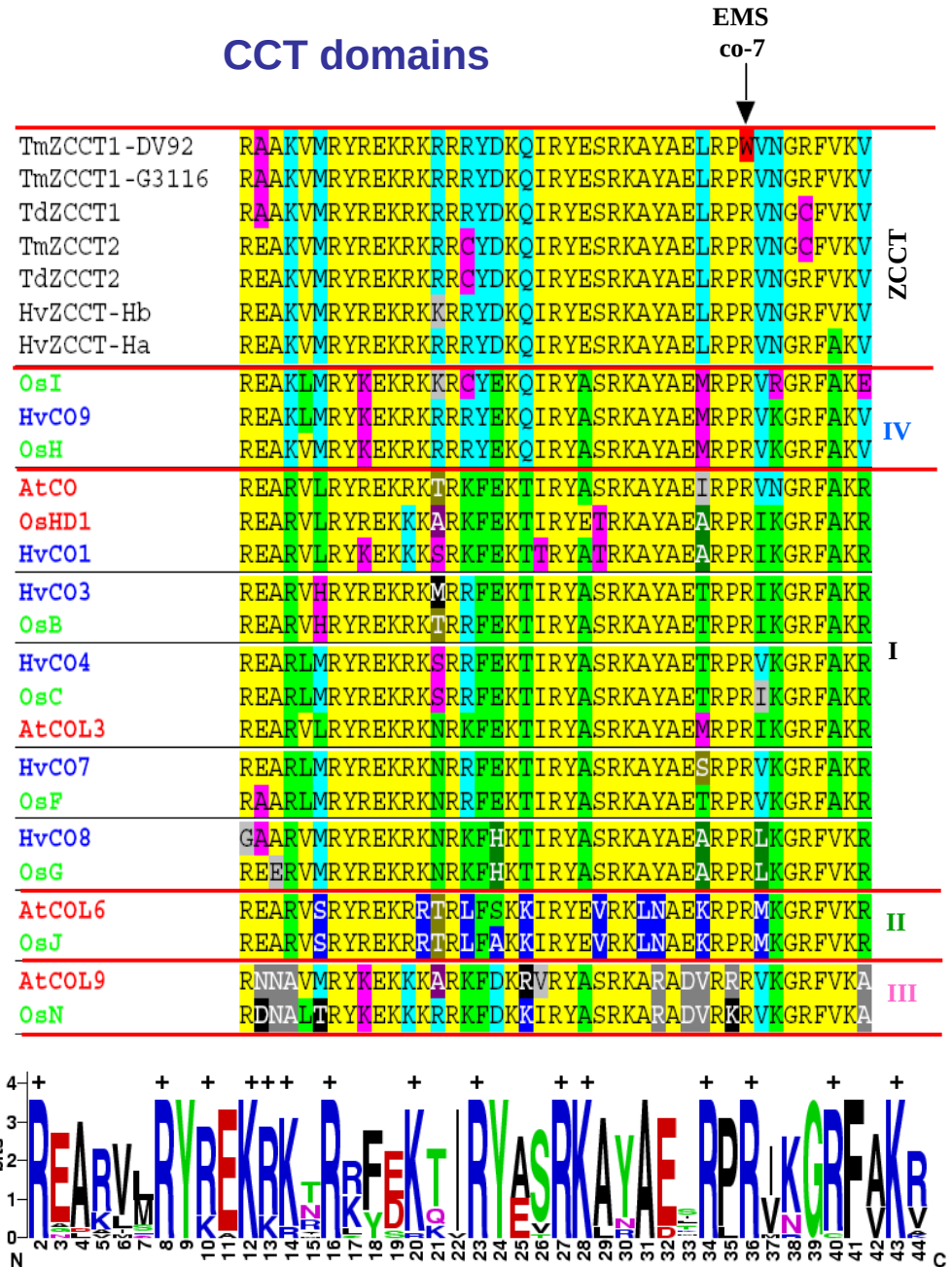
Yan et al. 2004. Science 303:1640



Cluster analysis of the CCT domains



CCT domains



Multiple Sequence Alignment (MSA)

1. Uses of MSA

2. Technical difficulties

1. Select sequences

2. Select objective function

3. Optimize the objective function

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2. Progressive algorithms

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Sequence selection

Past problem: not enough related sequences.

Today problem: too many related sequences!

- Use database searching to identify related proteins.
- Select **homologous** sequences.
- If sequences are similar along all the sequence use **Global MSA programs**.
- If not, PSI-Blast can help to define conserved regions.
- Or use **local MSA programs** (Gibbs sampler, Match-Box, or MACAW)
- Select a “representative” group of sequences. Large number of sequences are slow to compute and hard to analyze.
- The inclusion of a large number of a particular group of closely related proteins will dominate the alignment or profile (in spite of weighting schemes).

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MSA. Technical difficulties

Problem: Optimal pairwise alignment $\neq$ optimal multiple alignment

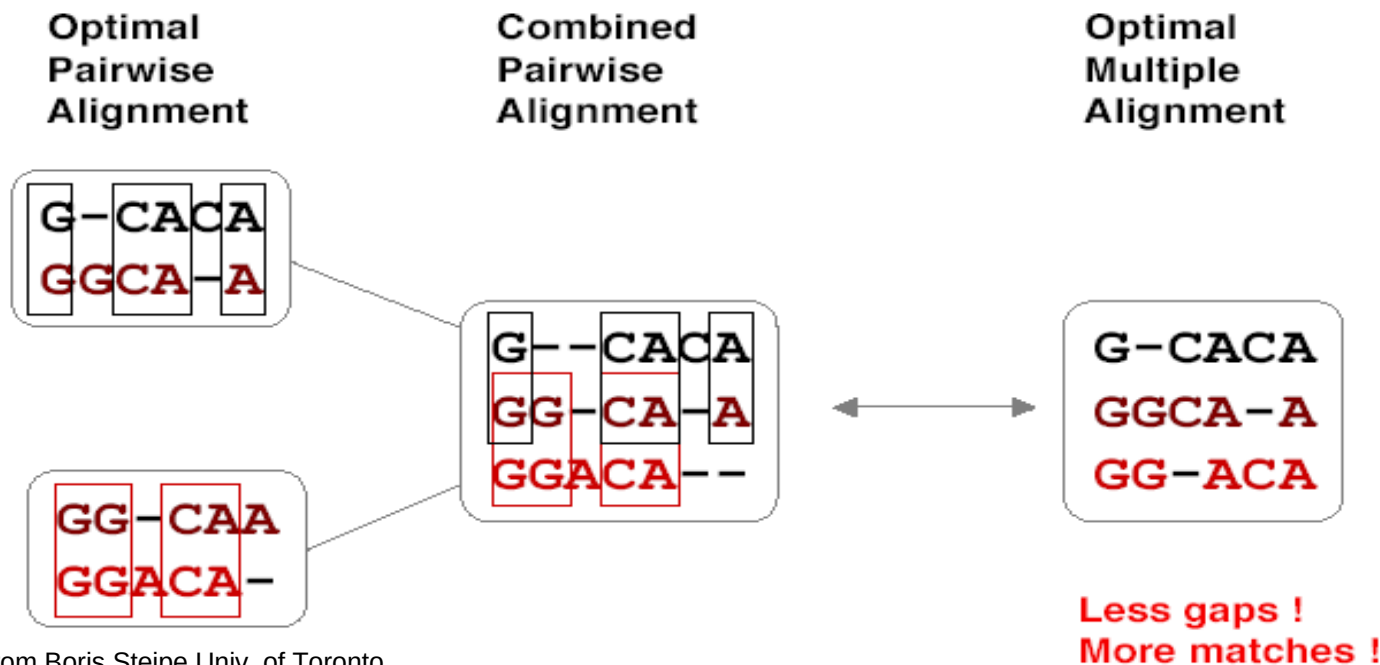


Fig. from Boris Steipe Univ. of Toronto

We need an Objective Function (OF) to measure which alignment is better

Objective functions (OF)

Define the mathematical objective of the search

Most widely used MSA packages use the OF “sum-of-pairs”

- Define a mathematical optimum
- Use sum of scores from pairwise alignments and affine gaps
- Use a Mutation Data Matrix (e.g. Blosum 62)
- Some add weighting proportional to the information in the seq.

A biologically ideal OF should

- Maximize similarity
- Minimize the number of gaps (over their length)
- Retain conserved motifs and patterns
- Retain functionally important alignments
- Recapitulate phylogeny
- Concentrate on alignable regions, not in gapped regions
- Consider the limitations imposed by the 3D structure

It is a non-trivial task to test the biological correctness of an objective function.

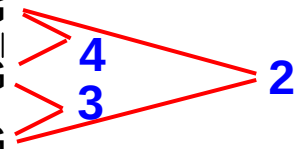
Simplest OF: Sum-of-pairs (SP)

After the best MSA is obtained, non-alignable sequences and spaces facing spaces are removed and a score is calculated for the **induced MAS** using any chosen scoring scheme (distance or similarity).

Seq. 1	AT-AATG	Induced Seq. 1-3 alignment		
Seq. 2	CTGAG-G	Seq. 1	AT-AATG	
Seq. 3	ATGAA-G	Seq. 3	ATGAA-G	
				Distance scheme: # mismatches (including -)

Sum-of-pairs score: The SP of a MSA is the sum of the scores of all the scores of the **induced** pairwise global alignments

Seq. 1 AT-AATG
Seq. 2 CTGAG-G
Seq. 3 ATGAA-G



Sum-of-pairs distance = 4 + 3 + 2 = 9

Weighted Sum-of-pairs score: each score can be multiplied by a weight to reflect evolutionary distances.

New Objective Functions

Multiple MSA: Depending on the Mutation Data matrix selected and on the selected gap penalties (opening and extension) very different MSA will be obtained. **Which one is the correct one?**

Seq.1	AT-AATG		Seq.1	ATAATG	Clustal
Seq.2	CTGAG-G	Distance scheme	Seq.2	CTGAGG	Gap open= 11
Seq.3	ATGAA-G	# mismatches (including -)	Seq.3	ATGAAG	Gap ext.= 1

New generation of Objective functions: less sensitive to gap penalty estimations thanks to the incorporation of local information

- **Segment-to-segment comparisons** of the sequences (instead of character-to-character) **without gap penalties** is the strategy used by **DiAlign**. This approach is efficient for **local similarities**, (genomic DNA, many protein families) <http://bibiserv.techfak.uni-bielefeld.de/dialign/>.
- **Consistency objective function**: (e.g. **T-Coffee** & **DiAlign**). The optimal MSA is defined as the one that agrees the most with all the optimal pair-wise alignments. "Given a set of independent observations the most consistent are often closer to "the truth".

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MSA. Technical difficulties

Problem: Optimal multiple alignment is intractable for $n \gg 2$

Pairwise alignment: $O(L^2)$ → 2D-Pathmatrix

n -wise alignment: $O(L^n)$ → n D-Path-hypercube

E.g. 10 GlnRS sequences – $L = 200$, $n = 10$: **10^{23} operations**

Despite more than thirty years of history, multiple sequence alignment is still a topic of ongoing research in bioinformatics.

MSA: Exact algorithm

MSA program

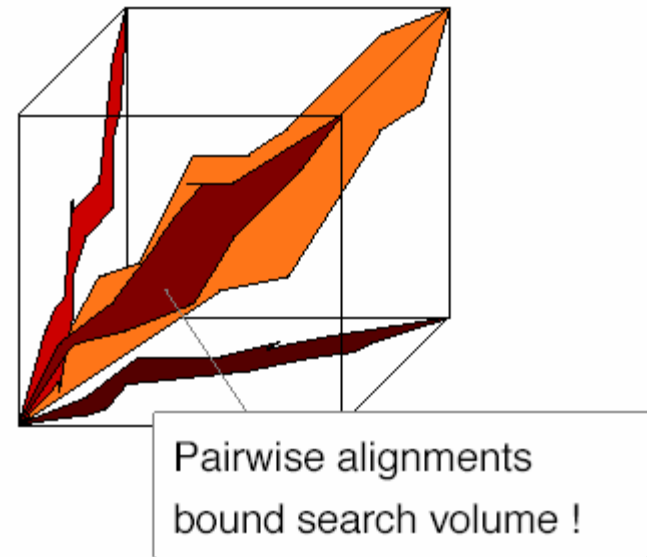
- Multidimensional dynamic programming
- Optimizes sum-of-pairs
- More accurate than progressive methods
- BUT... Time proportional to L^n
- Practical to ~10 seq. of $L < 200$ -bp

DCA (Dive & Conquer Algorithm)

- Sits on top of MSA program
- Produces simultaneous MSA
- Cuts seq. in subsets, that are fed into MSA
- Practical to ~20-30 seq. of $L < 200$ -bp
- Easy WEB submission

OMA (Optimal Multiple Alignment)

- Iterative implementation of DCA
- Speeds up DCA
- Decreases memory requirements



<http://bibiserv.techfak.uni-bielefeld.de/dca/>

<http://bibiserv.techfak.uni-bielefeld.de/oma/>

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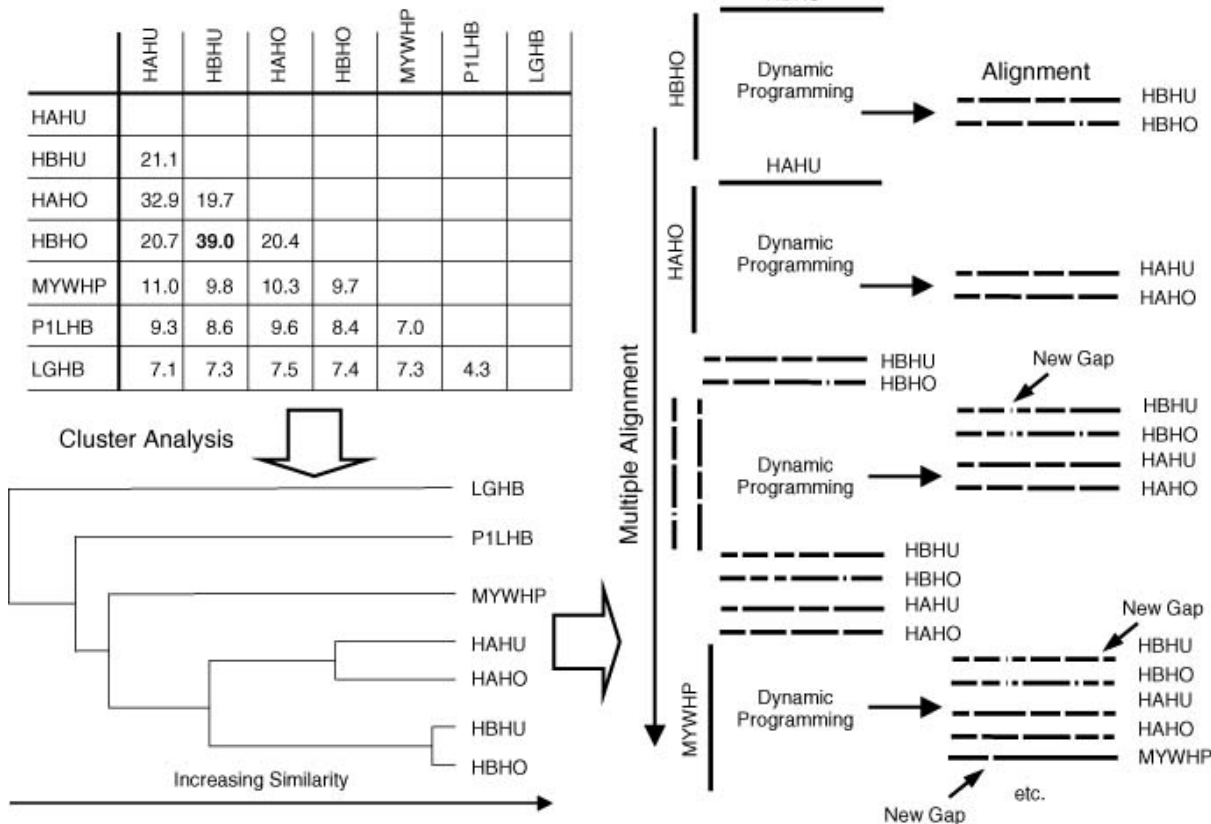
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Progressive algorithms (ClustalW, MultAlign, AMPS)



CLUSTALW

- Non-iterative & deterministic.
- OF: weighted sum-of-pairs.
- Affine gap penalties.
- Automatic substitution matrix choice.
- Most popular.
- Performs well in dense trees without obvious outliers (needs stepping stones).
- Can use SwissProt secondary structure information for gap penalty estimation.

Example of Progressive algorithm

- Calculate distances/similarities between sequences
- Construct a tree
- Add sequentially, following tree

Problems with progressive algorithms

1. Bad decisions taken in the initial alignments will persist throughout the process and cannot be corrected
2. For numerous sequences may take long time to calculate tree. MUSCLE is a good rapid alternative

DCA alignment

```
-----THEFA-----TCAT
GARFIELDTHEVERYFASTCAT
GARFIELDTHEFAS----TCAT
GARFIELDTHELASTFA-TCAT
```

```
-----THEFAT-----CAT
GARFIELDTHEVERYFASTCAT
GARFIELDTHEFASTCAT----
GARFIELDTHELASTFAT-CAT
```

ClustalW Blosom62 Gap 11-1
Cheaper to open distal gap
than to align C and F

```
GARFIELDTHEFASTCAT---
GARFIELDTHELASTFATCAT
```

```
GARFIELDTHEVERYFASTCAT
GARFIELDTHEFASTCAT----
GARFIELDTHELASTFAT-CAT
```

GARFIELD THE LAST FAST CAT

GARFIELD THE FAST CAT

GARFIELD THE VERY FAST CAT

THE FAT CAT

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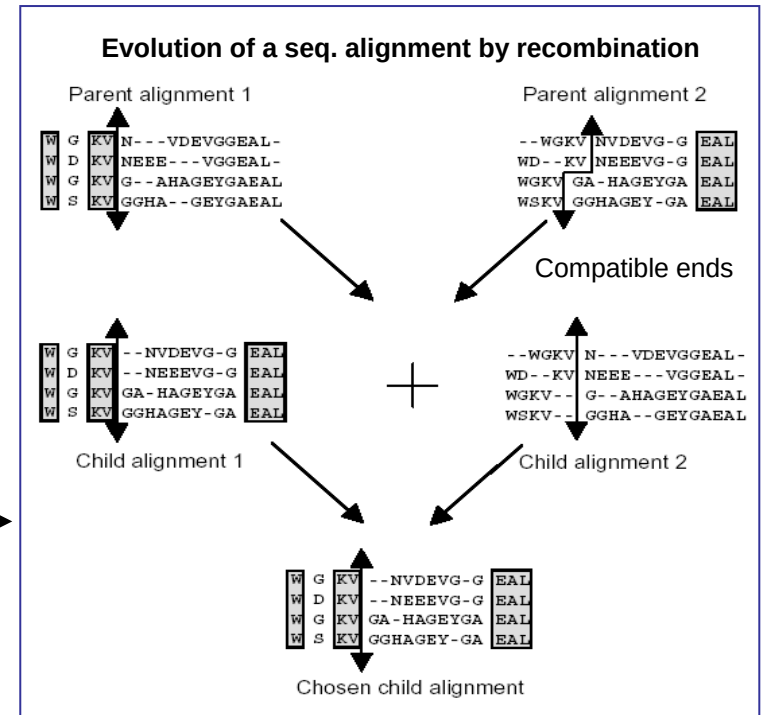
2. BOXSHADE & Seq. LOGOS

Iterative algorithms

Recurrent modifications of suboptimal solutions

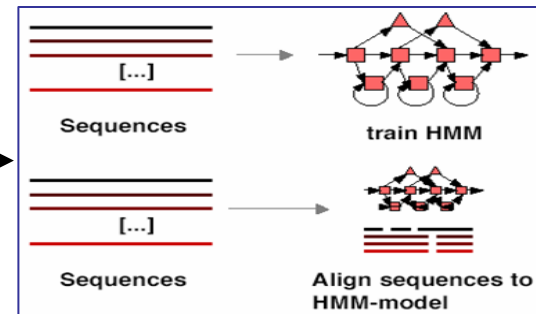
SAGA

- Uses a 'Genetic Algorithm'
- Can use different objective functions (e.g. Coffee)
- Mutations randomly insertion or shift gaps
- Sequences can recombine
- Sequences evolve, higher OF scores survive



Hammer (also SAM)

- Build Hidden Markov Models based on seq.
- Align sequences to HMM
- <http://hmmer.janelia.org/>



Gibbs sampler (Local MSA)

- Finds un-gapped motifs
- Segments are removed or added to increase a P value
- <http://bayesweb.wadsworth.org/gibbs/gibbs.html>

GAs and **HMMs** have been rather disappointing in *ab initio* alignments. Better: Pre-compute MSA with other program and then use this ones for optimization

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DiAlign Consistency-based algorithm

<http://bibiserv.techfak.uni-bielefeld.de/dialign/>

<http://bibiserv.techfak.uni-bielefeld.de/altavist/submission1.html>

<http://dialign-tx.gobics.de/index> (DiAlign-TX, 2008 improvement)

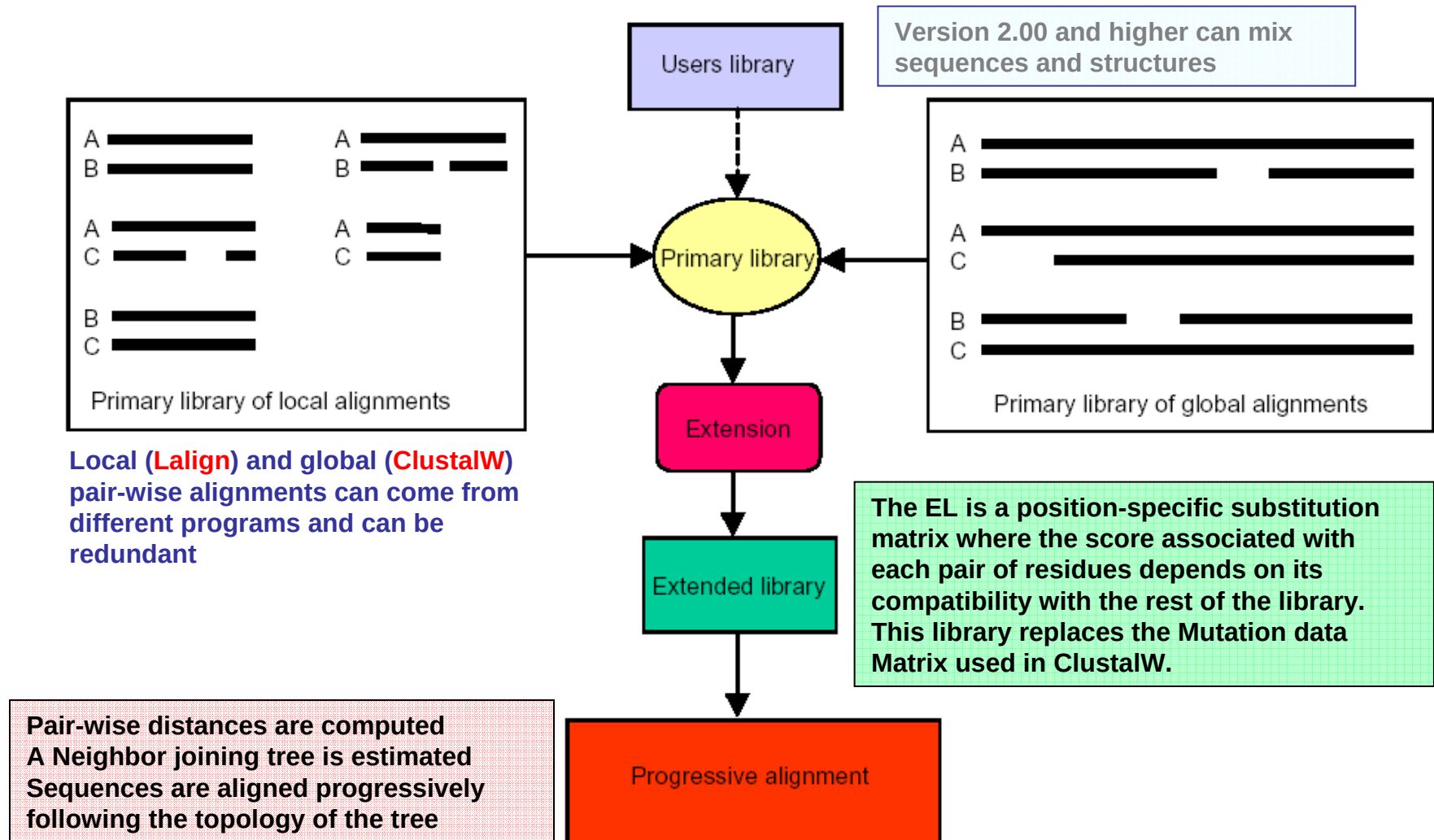
- 1.- Consistency based algorithm.
- 2.- Constructs pairwise and multiple alignments by comparing whole *segments* of the sequences rather than individual residues.
- 3.- No gap penalty is used.
- 4.- Efficient where sequences are not *globally* related but share only *local* similarities.

METHOD

- 1.- Starts with identification of highly similar segment pairs
- 2.- Each pair is weighted by its P values (similar to BLAST) and by a second score based on its compatibility with the complete set of segment pairs.
- 3.- The MSA is progressively assembled by adding the pairs of segments according to their weight

T-COFFEE: consistency-based algorithm

T-Coffee (Consistency **O**bjective **F**unction **F**or alignmEnt **E**valuation)



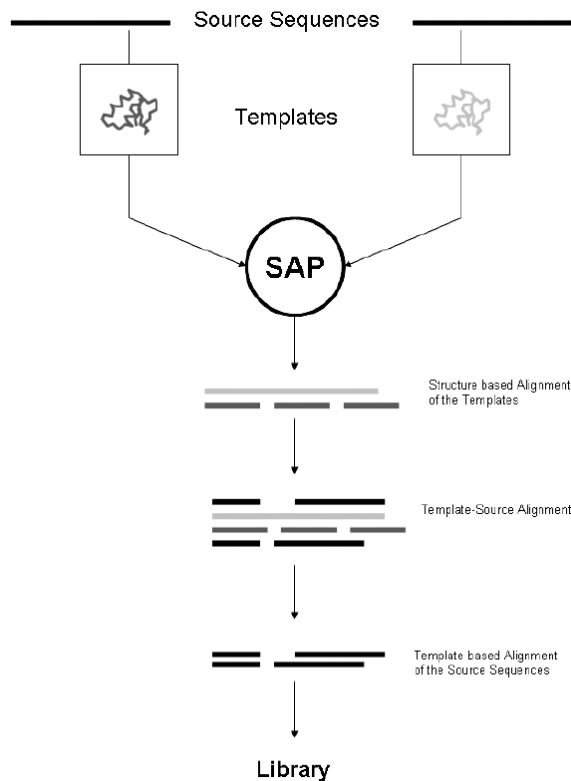
This figure indicates the layout of T-Coffee. Local and global pairwise alignments are first computed and then combined into a primary library that is extended in order to be used for computing the multiple sequence alignment in a progressive manner.

3D-COFFEE (Espresso): incorporates structures

<http://igs-server.cnrs-mrs.fr/Tcoffee/>

NAR. 2006 34:605-608

Espresso is a multiple sequence alignment server that aligns sequences using structural information. The user provide sequences and the server runs BLAST to identify close homologues of the sequences within the PDB structure database. Good for sequences with low similarity because STRUCTURE EVOLVES SLOWER THEN SEQUENCE!



- 1.- BLAST PDB structures (>60% identity, >70% coverage)
- 2.- LIBRARY CONSTRUCTION
 - Align sequences using lalign
 - Align templates to structures using SAP
 - Align structure induced alignments

The **M-Coffee** web server: a meta-method for computing multiple sequence alignments by combining alternative alignment methods

<http://www.tcoffee.org/>

PDB 10/06= 39,204 structures

<http://www.rcsb.org/pdb/>

Benchmark tests (from Notredame 2002)

Table 2. Some elements of validation on BALiBASE.

Method	Ref1	Ref2	Ref3	Ref4	Ref5	Total
DiAlign	71.0	25.2	35.1	74.7	80.4	57.3
ClustalW	78.5	32.2	42.5	65.7	74.3	58.7
T-Coffee	80.7	37.3	52.9	83.2	88.7	68.7

BALiBASE reference database

http://bips.u-strasbg.fr/fr/Products/Databases/BALiBASE/prog_scores.html

Ref1: homogenous set of sequences

Ref2: homogenous group of sequences and an out-layer

Ref3: contains two distantly related groups of sequences.

Ref4: contains sequences that require long internal gaps

Ref5: contains sequences that require long-terminal gaps

ClustalW: performed well on Ref. sets 1-3, but poorly on 4-5 when long internal or terminal gaps are required.

When large gaps required **T-Coffee** and **DiAlign** perform better

Summary strategies

Strategies for multiple alignment algorithms reflect biological heuristics

- **Exact** Bounded optimal solution search:
"Maximize similarity, Minimize gaps"
- **Progressive** Align most similar first, then add together
"Align according to phylogeny"
- **Consistency based** Conserved regions guide alignment
"Retain conserved regions"
- **+/- Iterated** Improve alignment from draft alignments
"Conserved regions of significance can be extended"
- **Combinations** Empirical combinations of above

From Boris Steipe Univ. of Toronto

Alignment Visualization

Alignment Visualization: e.g. BOXSHADE

Web server or local installation ...

MSF or CLUSTAL ALN format ...

Absolutely Conserved

Mostly Conserved

Somewhat Conserved

```
NYEU  MGLSTCEMGLVNLNVGKVEADIPGRCQVLTLEFRGHDPTELEKFDPRFRIKSDRMKASE
NYCS  -GLSTCEMGLVNLNVGKVEADIPGRCQVLTLEFRGHDPTELEKFDPRFRIKSDRMKASE
NYGO  -GLSTCEMGLVNLNVGKVEADIPGRCQVLTLEFRGHDPTELEKFDPRFRIKSDRMKASE
NYFC  -GLSTCEMGLVNLNVGKVEADIPGRCQVLTLEFRGHDPTELEKFDPRFRIKSDRMKASE
NYWER -VLSIAEWELVNLNVGKVEADIPGRCQVLTLEFRGHDPTELEKFDPRFRIKSDRMKASE
NYFN  -GLMIGEWQVLTLEFRGHDPTELEKFDPRFRIKSDRMKASE
NYAQ  MRLSTCEMGLVNLNVGKVEADIPGRCQVLTLEFRGHDPTELEKFDPRFRIKSDRMKASE
NYREJ -----TEWERVNRVAVVEIPAVOLAELREFRGHDPTELEKFDPRFRIKSDRMKASE
NYTUY -----ALFDIAVLKCGPPVADITMGGLVLTLEFRGHDPTELEKFDPRFRIKSDRMKASE
consensus -gladgacqlVlnvGkVtadi-ghGgavliRLfkgRpRTLeK7dRfkhkksadankase

NYEU  DLKKEGATVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
NYCS  DLKKEGATVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
NYGO  DLKKEGATVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
NYFC  DLKKEGATVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
NYWER DLKKEGATVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
NYFN  DLKKEGATVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
NYAQ  KMKQEGTVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
NYREJ DLKKEGATVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
NYTUY AASAGATVLTALGGLKKEGHEAEIRPQAQSLATKRRIPVYCLEFISECTIQVILGSKK
consensus dLkKkG-TVltALGGLKkKGbBeseLkpLaqSLatKkKIPVkyLefIse-ii-VI-skk

NYEU  PGDPGATAGAMNRALELFEFMASWKEELGFGG
NYCS  PGDPGATAGAMNRALELFEFMASWKEELGFGG
NYGO  PGDPGATAGAMNRALELFEFMASWKEELGFGG
NYFC  PGDPGATAGAMNRALELFEFMASWKEELGFGG
NYWER PGDPGATAGAMNRALELFEFMASWKEELGFGG
NYFN  ASWPGATAGAMNRALELFEFMASWKEELGFGG
NYAQ  PADPGATAGAMNRALELFEFMASWKEELGFGG
NYREJ PDMTGPMCESPSRVFTVICSITETLYKKEANFGG
NYTUY GLDACC--GTALRNVMGIIADLEANKEELGFGG
consensus pgdPgadAGam-kalelfr-Dwam-YKElgfgG
```

BOXSHADE: http://www.ch.embnet.org/software/BOX_form.html
<http://bioweb.pasteur.fr/seqanal/interfaces/boxshade.html>

http://www.ch.embnet.org/software/BOX_form.html

Multiple Sequence Alignment

Representation of multiple alignments: the consensus sequence

```
DIVMTQSPSSL...  
DIVMSQSPSSL...  
DIVMTQSPFTL...  
DVVMTQTPTLL...  
NIVLTQSPASL...  
DIQMTQSTSSL...
```

DIVMTQSPSSL...

"Consensus sequence"

BUT ... this is lossy.

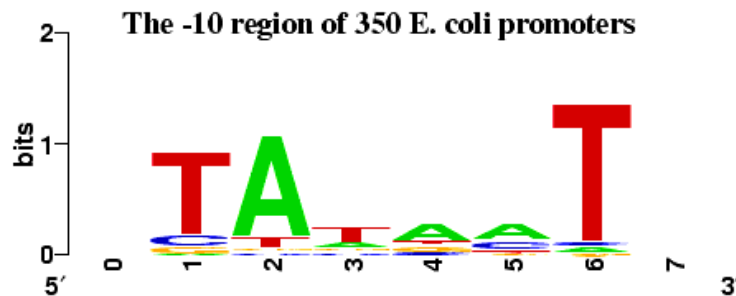
consider e.g. the TATAAT box:

49% **TAT...** 58% 54%

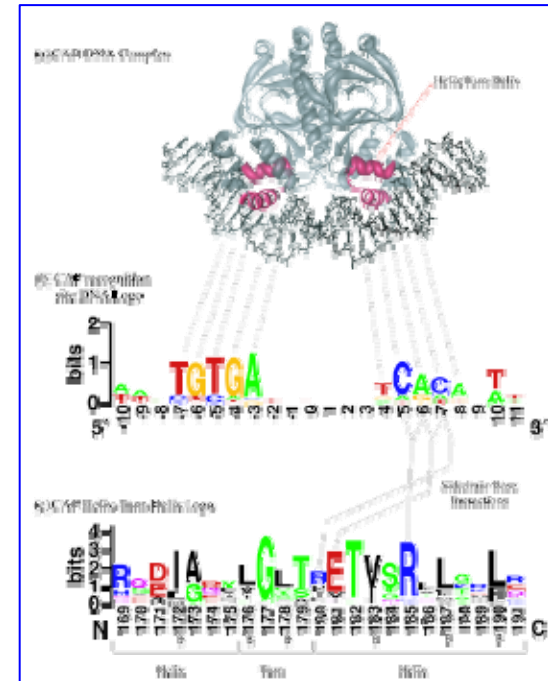
Only ~5% of bacterial promoters actually have the sequence TATAAT.

Representation of MSA

Sequence logos are a graphical representation of a MSA. Each logo consists of stacks of symbols, one stack for each position in the sequence. The height of symbols within the stack indicates the relative frequency of each amino or nucleic acid at that position.



WebLogo: application to generate **sequence logos**
<http://weblogo.berkeley.edu/>



Multiple Sequence Alignment

At the end of the day ...

- Use more than one alignment method,
- Make sure you have the right sequences.
- Don't align parts of sequences that can't be aligned (because they are not homologous).
- Realize problems from multi-domain proteins.
- Above all, use your common sense.